METRIC

DOD-R-18708B(NAVY)
31 August 1979
SUPERSEDING
MIL-R-18708A(NAVY)
11 October 1956
(See 6.4)

MILITARY SPECIFICATION

RUST-ARRESTING COATING, SHIPBOARD EXTERIOR,

GRAY, NAVY FORMULA NO. 123G (METRIC)

This specification is approved for use by all interested Commands of the Department of the Navy and is available for use by all other Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rust-arresting compound which will temporarily arrest ferrous metal corrosion and remove some rust until proper surface preparation and painting can be accomplished. This product may be used where air pollution regulations apply.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-L-1155 - Linseed Oil, Alkali Refined.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY

MIL-M-15205 - Metal Conditioning Compound.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials.
FED-STD-595 - Colors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 93 - Flash Point by Pensky-Martens Closed Tester, Test for.
D 209 - Lampblack, Specification for.
D 562 - Consistency of Paints Using the Stormer Viscosimeter, Test for.
D 600 - Liquid Paint Driers, Specification for.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT
Rules and Regulations
Rule 102

(Application for copies should be addressed to the South Coast Air Quality Management District, Metropolitan Zone, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Manufacture. The ingredients shall be mixed and ground as required to produce a product which is uniform and free from grit.

3.2 Composition. The rust arresting compound shall consist of ingredients conforming to the requirements of the applicable specifications and in the proportions shown in table I. The formula procedures is slightly in excess of 833 liters (100 gallons) to allow for normal manufacturing loss.

TABLE I. Formula No. 123G.

Ingredients	Specifications	Mass units per unit volume
Titanium dioxide	1/	170 2/
Lampblack	ASTM D 209	6
Linseed oil	TT-L-1155	195
Metal-conditioning compound	MIL-M-15205, type II	565
Lead drier	ASTM D 600, Class B	4.8
Manganese drier	ASTM D 600, Class B	2.0

1/ The titanium dioxide pigment shall be an aluminum oxide, modified chalk-resistant rutile, titanium dioxide containing no less than 90 percent titanium dioxide and no less than 1.8 percent aluminum oxide. Silicon dioxide and zinc oxide may also be present.

2/ The use of mass units in pounds results in a volume of approximately 100 gallons and kilograms results in a volume of approximately 833 liters.

3.2.1 The metal conditioning compound MIL-M-15205, type II, used in the manufacture of this rust arresting compound shall be an approved compound and shall be source inspected prior to use.

3.3 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.4). Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.4 Mass per unit volume. The mass per unit volume shall be not less than 1.08 kg per liter (9.0 pounds per gallon) and not more than 1.13 kg per liter (9.4 pounds per gallon) when tested as specified in 4.3.

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3.5 Consistency. The consistency determined by a Stormer Viscosimeter operating at 200 revolutions per minute shall be not less than 68 grams (g) (52 Krebs units) and not more than 104 g (62 Krebs units) when tested as specified in 4.3.

3.6 Color. The color, when applied over an absorbent (rust) surface, shall approximately match color 26270 of FED-STD-595.

3.7 Flash point. The flash point shall be above 93°C (200°F) when tested as specified in 4.3.

3.8 Titanium dioxide. The rust-arresting compound shall contain no less than 16.0 percent by mass titanium dioxide when tested as specified in 4.3.

3.9 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form 1813 found in FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and testing. Sampling, inspection, and testing shall be in accordance with method 1031 of FED-STD-141.

4.3 Finished coating. The coating shall be tested in accordance with the applicable methods of FED-STD-141 specified in table II and as hereinafter specified. Failure of any sample to pass any test and compliance with requirements of this specification shall be cause for rejection of the lot represented by the sample.

TABLE II. Test Methods.

Item	Applicable method in FED-STD-141	Applicable method in ASTM
Mass per unit volume		D 1475
Consistency		D 562
Flash point		D 93
Titanium dioxide	7081, 7082, or 7083	------

4.4 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Navy Bureau of Medicine and Surgery, Department of the Navy, Washington, DC 20372. The disclosure of proprietary information, which will be held in confidence by the Bureau of Medicine and Surgery, shall include: the name, formula, and approximate percentage by mass and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material.

4.5 Inspection of preparation for delivery. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging, packing, and marking. The coating shall be furnished in 4-liter or 1-gallon cans or 20-liter or 5-gallon pails, as specified (see 6.2). The coating shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging and packing A, B, or C, respectively shall be as specified (see 6.2).

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5.1.1 Special marking. In addition to other markings required on the containers, there shall be the following statement: "The volatile content of this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District".

6. NOTES

6.1 Intended use. The rust-arresting compound covered by this specification is intended for use on external, ferrous metal, weather surfaces of ships whenever temporary protection against corrosion is needed and proper surface preparation and painting cannot be accomplished.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and level of packing required (see 5.1).
(c) Size of container required (see 5.1).

6.3 Unit quantities. The compound should be purchased by volume, the unit being 1 liter or 1-gallon (231 cubic inches) at 15.6°C (60°F).

6.4 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-R-18708B(N) METRIC

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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